

1. What is the difference between frontend, backend, and full-stack development?

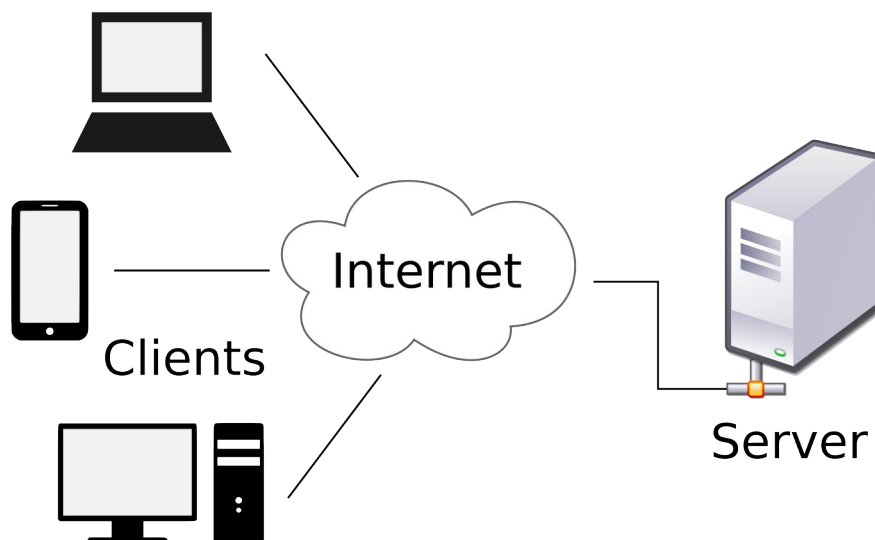
Answer:

Frontend development involves building the user interface and user experience of a website or application, focusing on what users see and interact with. Backend development deals with server-side logic, databases, and application functionality that users do not see. Full-stack development combines both frontend and backend skills to handle the entire application development process.

Example: Frontend — designing a website's layout with HTML, CSS, JavaScript; Backend — handling user authentication and database transactions; Full-stack — developing and maintaining both the frontend and backend parts.

2. Draw a simple diagram showing how the client-server model works in web architecture.

Answer:



3. How does a browser request and display a web page from a web server?

Answer:

A browser sends an HTTP request to a web server for a specific page URL. The server processes the request, retrieves the page resources, and sends back an HTTP response with the HTML, CSS, and JavaScript files. The browser then renders the HTML, applies CSS for styling, and executes JavaScript for interactivity, displaying the web page to the user.

4. What tools are required to set up a web development environment? Explain the purpose of each.

Answer:

Code Editor (e.g., VS Code): To write and edit code.

Web Browser (e.g., Chrome, Firefox): To test and debug web pages.

Version Control System (e.g., Git): To manage code changes.

Terminal/Command Line Interface: To run commands and scripts.

Package Managers (e.g., npm): To manage libraries and dependencies.

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5. What is a web server? Give examples of commonly used servers.

Answer:

A web server is software that handles HTTP requests from clients (browsers) and serves web content. Examples include Apache HTTP Server, Nginx, and Microsoft IIS.

6. What are the roles of a frontend developer, backend developer, and database administrator in a project?

Answer:

Frontend Developer: Designs and implements the user interface.

Backend Developer: Builds the server-side logic and APIs.

Database Administrator (DBA): Manages the database system, ensuring data integrity and security.

7. How do you install VS Code and configure it for HTML, CSS, and JavaScript development?

Answer:

Download VS Code from its official site, install recommended extensions such as Live Server for real-time preview, and set up settings.json for formatting and linting. (Screenshot task is not feasible here).

8. What is the difference between static and dynamic websites? Provide an example of each.

Answer:

Static websites deliver fixed content to users where pages are pre-built and do not change per user request (e.g., a company's brochure site). Dynamic websites generate content on the fly based on user interaction or backend logic (e.g., social media sites like Facebook).

9. List five web browsers. How do rendering engines differ between them?

Answer:

Five web browsers: Google Chrome, Mozilla Firefox, Microsoft Edge, Safari, Opera. Rendering engines differ as Chrome and Edge use Blink, Firefox uses Gecko, Safari uses WebKit, and Opera uses Blink as well. Each engine interprets HTML, CSS, and JavaScript differently, affecting page rendering speeds and compatibility.

10. Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

Answer:

